

Enterprise Data Engineering, Analytics & AI

ETL vs ELT

Batch vs Real-Time & Industry Tools Overview

Data Engineering Fundamentals — Conceptual Module

Today's Agenda

60-Minute Roadmap

01

What is Data Pipeline?

Foundation + Quick Recap

02

ETL — Extract, Transform, Load

Concept · Flow · Real Examples

03

ELT — Extract, Load, Transform

Concept · Flow · Real Examples

04

ETL vs ELT — Head to Head

When to use what & Why

05

Batch vs Real-Time Processing

Definitions, Trade-offs, Examples

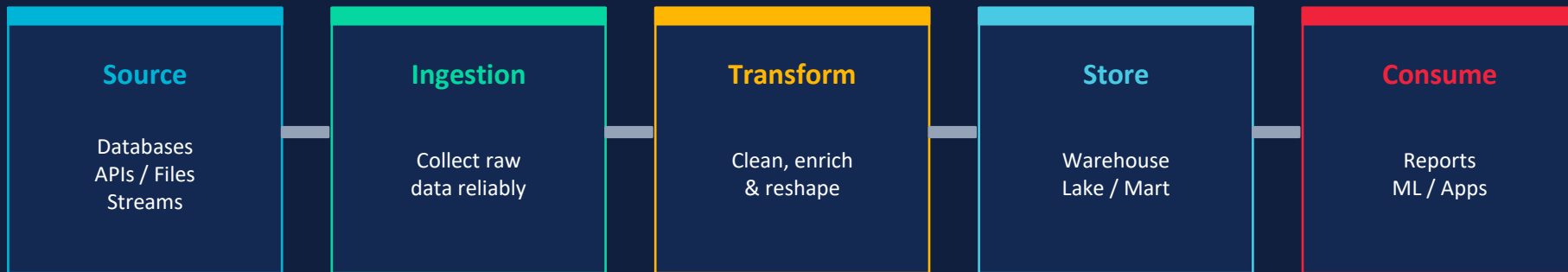
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Industry Tools & Platforms

Popular tools mapped to use cases

What is a Data Pipeline?

Foundation concept before diving into ETL & ELT



Real-World Analogy:

A pizza restaurant pipeline: Raw ingredients arrive (Source) → Chef preps them (Transform) → Stored in fridge (Store) → Served to customer (Consume). A data pipeline does the same — but with data!

Key insight: ETL & ELT are two different approaches to managing this pipeline.

ETL — Extract, Transform, Load

The Traditional Approach — Processing data BEFORE storing it

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EXTRACT

Pull raw data from multiple sources

- Databases (MySQL, Oracle)
- REST APIs & Web Services
- CSV / Excel files
- Legacy systems

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TRANSFORM

Clean & reshape data in a staging area

- Remove duplicates
- Validate data types
- Join / aggregate
- Apply business rules

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LOAD

Load clean data into the destination system

- Data Warehouse (Snowflake)
- Relational DB (PostgreSQL)
- OLAP Cubes
- Reporting systems

⚡ Key: Transformation happens OUTSIDE the target system — in a dedicated staging server or ETL tool, before loading.

ETL in Action — Real Example



Scenario: A Bank generating a nightly customer loan summary report

1. EXTRACT

Each night at 11 PM, pull transaction records from Oracle DB, account data from MySQL, and KYC data from a flat CSV file exported by the compliance team.

2. TRANSFORM

In an ETL server (e.g., Informatica): remove test/dummy accounts, convert all currencies to INR, calculate loan utilization %, flag high-risk accounts, join all three sources.

3. LOAD

Cleaned, structured data is loaded into the bank's central Data Warehouse (Teradata) — ready for the morning dashboard reports & risk team.

Result: Every morning, the risk dashboard shows clean, standardized, up-to-date loan data — made possible by overnight ETL.

ELT — Extract, Load, Transform

The Modern Approach — Load raw data FIRST, transform INSIDE the target system

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EXTRACT

Pull raw data from sources as-is

- Cloud APIs
- Event streams
- Databases
- SaaS platforms

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LOAD

Load RAW data directly into cloud storage / lake

- Amazon S3
- Azure Data Lake
- BigQuery raw tables
- Snowflake staging

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TRANSFORM

Transform inside the target using SQL / dbt

- SQL transforms in-DB
- dbt models
- Spark on cloud
- In-warehouse compute

⚡ Key: Raw data is stored first — transformation happens INSIDE the data warehouse using its own compute power (e.g., BigQuery, Snowflake).

ELT in Action — Real Example

 **Scenario: An E-Commerce Company (like Flipkart) doing real-time analytics on customer behaviour**

1. EXTRACT

Pull clickstream data from website, order data from microservices, and customer profile data from Salesforce CRM — all happening every few minutes via Fivetran or Airbyte.

2. LOAD

All raw, uncleaned data lands directly into Google BigQuery (Data Warehouse) as raw tables. Hundreds of GBs stored immediately — no waiting for transformation.

3. TRANSFORM

dbt (data build tool) runs SQL models inside BigQuery: calculates conversion rates, user cohorts, product performance — using BigQuery's massive compute. Analysts can query instantly.

Result: Analysts explore raw + transformed data without waiting — modern ELT enables this speed at massive scale.

ETL vs ELT — Head to Head

Criteria	ETL	ELT
Transform Location	External staging server	Inside the data warehouse
Data Volume	Best for small-medium data	Built for big data / cloud scale
Schema Requirement	Schema-on-write (predefined)	Schema-on-read (flexible)
Speed to Load	Slower (transform first)	Faster (load raw immediately)
Raw Data Access	✗ Not preserved	✓ Raw data always accessible
Cost	ETL server + license costs	Pay for cloud compute
Best For	Banking, Compliance, OLAP	Analytics, ML, Modern cloud

When to Use ETL vs ELT?

Choose the right approach based on your context

✓ USE ETL WHEN...

- Data must be clean BEFORE storing (compliance)
- Working with legacy on-premise systems
- Strict data governance & security requirements
- Small/medium data volumes (< TB range)
- Industry: Banking, Healthcare, Insurance
- Tools: Informatica, SSIS, Talend

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✓ USE ELT WHEN...

- Using cloud data warehouses (BigQuery, Snowflake)
- Working with large or unstructured datasets
- Analysts need raw data access for exploration
- Speed of ingestion is more important
- Industry: E-Commerce, SaaS, Tech companies
- Tools: dbt, Fivetran, Airbyte + BigQuery/Snowflake

Batch vs Real-Time Processing

Two fundamentally different philosophies of data processing

BATCH PROCESSING

What it is:

Collect data over time, then process it all at once in a scheduled job (e.g., nightly or hourly).

Analogy:



Like a bank that processes all transactions at end-of-day, not one-by-one in the moment.

When to use:

- High data volume, non-urgent reporting
- Monthly payroll / billing systems
- Nightly ML model retraining
- End-of-day reconciliation reports

REAL-TIME (STREAM) PROCESSING

What it is:

Process each data event the moment it arrives — continuous, low-latency computation.

Analogy:



Like a traffic signal system that reacts to car sensors instantly — not at the end of the day.

When to use:

- Fraud detection (credit card alerts)
- Live sports scores & dashboards
- IoT sensor monitoring
- Recommendation engines (Netflix/Swiggy)

Batch vs Real-Time — Comparison

Dimension	Batch	Real-Time
Data Freshness	Minutes to hours old	Milliseconds to seconds
Latency	High (scheduled jobs)	Very low (continuous)
Complexity	Simpler to build & debug	Complex, needs careful design
Cost	Lower compute cost	Higher continuous compute
Use Case Scale	Large historical datasets	Event-by-event processing
Tools	Spark, Hadoop, Airflow	Kafka, Flink, Spark Streaming
Example	Monthly sales report	Credit card fraud alert

Industry Tools & Platforms

Tools organized by category — Know what each one does

ETL Tools

- Informatica PowerCenter
- Microsoft SSIS
- IBM DataStage
- Talend Open Studio
- Pentaho

ELT / Ingestion

- Fivetran
- Airbyte (open source)
- Stitch
- dbt (Transform layer)
- AWS Glue

Data Warehouses

- Google BigQuery
- Snowflake
- Amazon Redshift
- Azure Synapse
- Databricks

Stream / Real-Time

- Apache Kafka
- Apache Flink
- Spark Streaming
- AWS Kinesis
- Google Pub/Sub

Orchestration

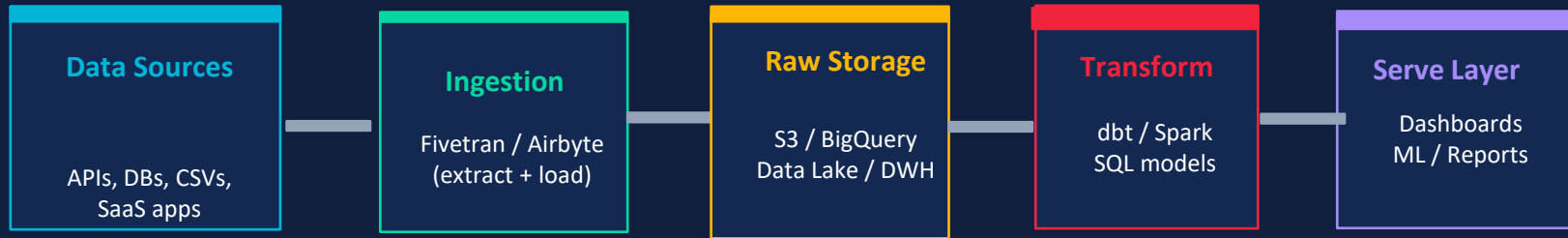
- Apache Airflow
- Prefect
- Dagster
- AWS Step Functions
- Luigi

Data Lakes

- Amazon S3 + Athena
- Azure Data Lake Storage
- Google Cloud Storage
- Delta Lake
- Apache Hudi

How Tools Fit Together — A Modern Data Stack

A typical cloud-native data pipeline (used by companies like Swiggy, PhonePe, etc.)



⚙️ **Orchestration Layer:** Apache Airflow / Prefect schedules and monitors all jobs above

⚡ **Real-Time Path:** Events → Apache Kafka → Spark Streaming / Flink → Dashboard (milliseconds latency)

ETL/ELT paradigm handles the BATCH path above. Real-Time is the STREAMING path below.

Key Takeaways

What you must remember from today's session

01

ETL transforms data BEFORE loading — used in legacy/on-premise, compliance-heavy industries

02

ELT loads raw data FIRST, then transforms INSIDE the warehouse — modern cloud-native approach

03

Use ETL for small/medium structured data with strict governance; ELT for big data & analytics

04

Batch processing = scheduled, high-volume, delayed; Real-Time = instant, event-driven, low-latency

05

The Modern Data Stack = Fivetran/Airbyte + Cloud DWH (Snowflake/BigQuery) + dbt + Airflow

06

Kafka powers real-time pipelines; Spark/Flink process the stream; Airflow orchestrates batch jobs

Questions & Discussion

Let's consolidate today's learning together

Think & Discuss:

1. Your bank sends you a fraud alert on your phone within 2 seconds of a transaction. Which processing type is this?
2. A hospital wants to generate monthly patient reports. Should they use ETL or ELT? Why?
3. Swiggy shows you a restaurant recommendation based on what you just browsed. Which pipeline pattern is involved?

Coming Up Next:

- W3: Data Modeling — Star Schema, Snowflake Schema & Dimensional Modeling
- W13: Hands-on ETL & ELT — Building your first pipeline (practical labs)
- Recommended: Explore dbt Fundamentals (free course at courses.getdbt.com)